Drug Decriminalization and College Graduation: Evidence from Oregon's Measure 110

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Abstract

On February 1, 2021, Oregon became the first state in the United States to decriminalize the possession of small quantities of all illicit drugs under Measure 110. This unprecedented policy shift has sparked extensive debate regarding its broader societal implications. Proponents assert that decriminalization can reduce stigma, encourage treatment for substance use disorders, and lessen the criminal justice burden. However, critics caution that such measures could inadvertently increase drug use and overdose rates. This paper constructs a synthetic Oregon, outlined by Abadie, Diamond, and Hainmueller (2010), to examine the causal impact of Measure 110 on college graduation rates in Oregon, offering insights into how drug policy changes may influence educational attainment.

1 Introduction

Does drug decriminalization reduce college graduation rates? One can imagine two plausible scenarios in which this policy might influence students' academic outcomes. First, for those who already use drugs, decriminalization—alongside increased social programs addressing substance use—may help them graduate on time by minimizing involvement with the criminal justice system and providing improved treatment options. Alternatively, the policy could raise overall drug use by reducing the legal repercussions of consuming or selling illicit substances, potentially interfering with students' ability to complete their degrees.

Historically, attending college has been considered a protective factor against the development of substance use disorders (Welsh, Shentu, & Sarvey, 2019). However, in a study by Caldeira et al. (2009), nearly half of 946 college students who were followed from freshman to junior year met criteria for at least one substance use disorder during that time. Countless studies have shown that students who regularly use substances are more likely to have lower GPAs, spend fewer hours studying, miss significantly more class time, and fail to graduate on time (Welsh et al., 2019).

College students occupy a unique position in the drug policy debate because they are at a critical transitional stage: many are experiencing newfound independence, encountering academic and social pressures, and establishing long-term habits that can influence their future well-being. This combination of factors makes them both a vulnerable population and a group with significant potential for positive outcomes if appropriate support structures and policies are in place.

Moreover, college graduates typically have higher lifetime earnings, lower unemployment rates, and are more likely to engage civically. Ensuring that students can remain enrolled and progress toward graduation—rather than dropping out due to legal issues, health problems, or financial strains associated with substance use—directly contributes to the economic and social welfare of the state. Consequently, when contemplating drug policy reforms, the potential impact on college students' academic success and subsequent professional opportunities becomes a vital consideration, as it links public health interventions to broader educational and economic goals.

Despite the importance of understanding how statewide drug policies affect educational outcomes, there appears to be no existing research examining Measure 110's impact on college graduation rates. This paper addresses that gap by providing the first causal estimates of how decriminalization under Measure 110 has influenced college completion in Oregon.

2 Institutional Background

Historically, U.S. drug policy has been largely rooted in prohibition and strict enforcement, highlighted by the "War on Drugs" launched in the 1970s by President Richard Nixon, resulting in widespread criminalization of drug-related activities and a surge in incarceration rates, often disproportionately affecting minority groups. Throughout the 1980s and 1990s, mandatory minimum sentencing laws reinforced this punitive model, raising concerns about its social and economic costs. By the late 20th and early 21st centuries, policymakers and health professionals increasingly questioned the efficacy of a purely punitive approach, pointing to issues of addiction, public health, and prison overcrowding. In response, some regions began experimenting with alternative frameworks—most notably Portugal in 2001, which decriminalized all illicit drugs and channeled resources toward treatment and harm reduction. Within the United States, Oregon's Measure 110 represents the first comprehensive statewide effort to decriminalize the possession of all drugs for personal use. Reflecting a broader shift toward viewing addiction as a public health challenge rather than a criminal one, such measures aim to reduce stigma, encourage treatment, and ultimately address the root causes of substance abuse.

2.1 Measure 110

Measure 110, passed in November 2020, was a law aimed at shifting the state's approach to substance use from criminal punishment to health-oriented treatment. Taking effect on February 1, 2021, it reduced penalties for possessing small amounts of certain controlled substances—such as cocaine, heroin, methamphetamine, and ecstasy—from felonies or misdemeanors to a new Class E violation, carrying a maximum fine of $100. Prior to this, possession of these substances was a Class A criminal misdemeanor punishable in Oregon by up to a year in prison and/or fines up to $6,250. Alongside this decriminalization, the law established Behavioral Health Resource Networks (BHRNs) aimed at providing treatment and support, funded by redirected marijuana tax revenue and law enforcement savings.[1] However, in 2024, drug possession was recriminalized following a campaign by opponents of the original bill who argued the measure had failed to address the drug crisis effectively. The Oregon legislature passed House Bill 4002 in 2024, rolling back parts of Measure 110.[2]

3 Literature Review

One of the primary concerns about changing drug policies is their potential impact on youth education and development, especially given the substantial body of public health and health economics research linking drug use to negative academic and longer-term life outcomes. Chatterji (2003) explores the relationship between adolescent drug use and educational attainment, finding a negative association between marijuana and cocaine use during high school and the number of years of schooling completed. Using state-level drug policies as instruments, Chatterji's findings suggest that substance use during critical educational years has a significant detrimental effect on long-term educational outcomes, reducing educational attainment by 0.2 to 0.4 years for those who used marijuana or cocaine during high school

[1] https://olis.oregonlegislature.gov/liz/2019I1/Downloads/CommitteeMeetingDocument/227319

[2] https://www.opb.org/article/2024/08/29/measure-110-drug-law-deflection-posession-crime-law-oregon-recriminalization-decriminalization/

(Chatterji, 2003). These findings are further reinforced by Arria, Caldeira, Bugbee, Vincent, and O'Grady (2015), which find, via a longitudinal cohort study and structural model, that marijuana use in college students is associated with various negative educational outcomes including more missed classes, lower GPA, and longer times to graduation. Similarly, Mezza and Buchinsky (2021) develop a dynamic structural model that integrates decisions on drug use, education, and labor force participation. Their findings indicate that early drug use, particularly before age 19, is associated with lower educational attainment and poorer labor market outcomes later in life. This study emphasizes the potential long-term ramifications of youth exposure to drugs, suggesting that increased access to substances due to decriminalization could hinder educational achievement by altering incentives to invest in schooling (Mezza & Buchinsky, 2021).

Health outcomes, particularly overdose rates, are a critical area of concern when evaluating the impact of drug decriminalization. Spencer (2023) examines the causal effect of Oregon's Measure 110 on drug overdose deaths using the synthetic control method. He finds that decriminalization led to a 23% increase in drug overdose deaths in Oregon during the first year post-implementation, representing 181 additional fatalities compared to a synthetic control group constructed from other U.S. states (Spencer, 2023). Contrasting these findings, Zoorob, Park, Kral, Lambdin, and Del Pozo (2024) present a more nuanced analysis that controls for the rapid introduction of fentanyl into Oregon's unregulated drug market. They find that the surge in fatal overdoses in Oregon coincided with the influx of fentanyl, making it difficult to attribute the increase solely to decriminalization. After accounting for fentanyl as a potential confounder, the effect of Measure 110 on overdose mortality becomes statistically insignificant (Zoorob et al., 2024). This suggests that, while the timing of Measure 110's implementation and increased overdoses were correlated, fentanyl's emergence may have been the primary driver, though whether Oregon had a more drastic increase in fentanyl supply compared to other states is unclear.

Because Measure 110 focused on harm reduction rather than punitive measures, it is crucial to situate this policy within the broader context of similar approaches. Maghsoudi, McDonald, and Thompson (2021) conduct a systematic review of drug checking services, which aim to provide chemical analysis of drugs to reduce harm among users. While these services have been primarily used in European contexts, findings indicate that providing users with accurate information about drug composition can alter consumption patterns and reduce risky behaviors, such as the use of contaminated substances (Maghsoudi et al., 2021). This suggests that complementary harm reduction strategies may be essential to mitigate the risks of decriminalization.

Although the literature consistently points to the negative effects of drug use and remains inconclusive regarding Measure 110's impact on overdoses, the key question is how a more lenient drug policy might influence drug use among youth and college-age populations. In Oregon specifically, Kerr, Bae, and Koval (2018) find that recreational marijuana laws (RMLs) did not significantly increase marijuana use among college students, though it is important to note that marijuana use of college-aged students may be endogenous to RMLs being passed. Wen, Hockenberry, and Cummings (2015), on the other hand, examine the effects of Medical Marijuana Laws (MMLs) on various substances. They find that while marijuana use increased among adults post-MML, there were no significant changes in alcohol or other drug use among youth, indicating that perceptions and enforcement play a crucial role in shaping policy outcomes (Wen et al., 2015).

Despite the existing body of research on drug decriminalization and its health and behavioral impacts, there is a gap in understanding how policies regarding 'harder' drugs may affect educational outcomes, particularly for college students. Given the significant controversies regarding Measure 110 and the potential for increased drug exposure in communities post-decriminalization, it is essential to explore whether these changes have created an environment that is conducive to college completion. By focusing on college graduation rates, this study aims to fill a critical gap in the literature and provide insights into how policy shifts in drug regulation may shape the educational trajectories of youth and young adults.

Understanding these dynamics is crucial for policymakers as they navigate the complex balance between public health, safety, and educational priorities in the context of drug policy reform.

4 Data

I use data from the National Center for Education Statistics (NCES), specifically the Integrated Postsecondary Education Data System (IPEDS), along with the American Community Survey (ACS) 1-year tables. IPEDS is a system of 12 interrelated survey components conducted annually that gathers data from every college, university, and technical and vocational institution that participates in the federal student financial aid programs. The data collections occur in fall, winter, and spring. This is the most comprehensive postsecondary, college-inclusive data collection available. Within the IPEDS survey, I gather institutional characteristic data, graduation rates, and enrollment figures at the institution level from 2012 to 2023. I then merge these with state-level economic data from the ACS for the same period, creating a master dataset from which I construct a synthetic Oregon.

Table 1: Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Graduation Rate	612	47.442	7.555	24.061	70.500
Enrollment Total	561	4546.794	1866.839	1883.741	16054.120
Grad Degree	561	61.489	36.403	8.300	93.900
Unemp. Rate	561	5.665	1.962	2.200	12.300
Median Income	561	62108.014	13282.897	37095.000	108210.000

Graduation Rate and Enrollment Total come from IPEDS. Enrollment Total was missing for 2023 since it had yet to be calculated. Grad Degree comes from the ACS. Its ACS series definition changes within the panel—"percent of population 25 or older that is a high school graduate or higher" through 2018, and "percent with a graduate or professional degree" from 2019 onward—which accounts for the wide range in Table 1; because only the 2012 and 2016 values (both on the earlier definition) enter the synthetic-control predictors, the estimation is unaffected. The remaining variables also come from the ACS. All ACS estimates were missing for 2020. No missing predictor variables are of concern when constructing synthetic Oregon since no pre-treatment data is missing.

I restrict my sample to institutions offering two-year or higher degrees. Table 1 shows pooled averages of my variable of interest, college graduation rates, as well as the predictors used to construct synthetic Oregon. Total enrollment is collected yearly at the institutional level, which I then collapse by year and state; thus the table shows the average total enrollment by institution by state.

Figure 1 shows the country-wide average graduation rate. The graduation rate is calculated by IPEDS by cohort—full-time, first-degree-seeking students in a particular year—that have completed their program within 150% of the normal time period. For a 4-year degree this would mean that the student completes their degree within 6 years of entering the program. For a specific year, say 2018, this is the cohort graduating in the spring of 2019. As seen in Figure 1, the country-wide graduation rate, for all institutions, has increased by about 5 percentage points. Even though Measure 110 was passed in November of 2020 and did not go into effect until February 2021, this is still our '2020' cohort.

Figure 2 shows Oregon's average graduation rate over time versus the rest of the country. Figures 3 and 4 split Figure 2 by public and private institutions. In Oregon the graduation rate for public institutions is lower on average than the rest of the country but is trending upwards along with the country, while private institutions in Oregon have higher graduation rates on average than the rest of the country but are much more volatile and seem to be decreasing. This difference motivates opportunities for later heterogeneity analysis. All figures show averages for both 2- and 4-year institutions.

5 Econometric Strategy

In this paper, I estimate the effect of Measure 110 on college graduation rates using a synthetic control analysis, a method well-suited to policies enacted in a single state. This approach involves constructing a "synthetic" Oregon from a donor pool consisting of the other 49 states and Washington, D.C., thereby allowing me to measure the Average Treatment Effect on the Treated (ATT)—that is, the difference in graduation rates with and without Measure 110. To do this, I select a set of predictors to weight the donor pool, mini-

Figure 1: Country-Wide Average Graduation Rate

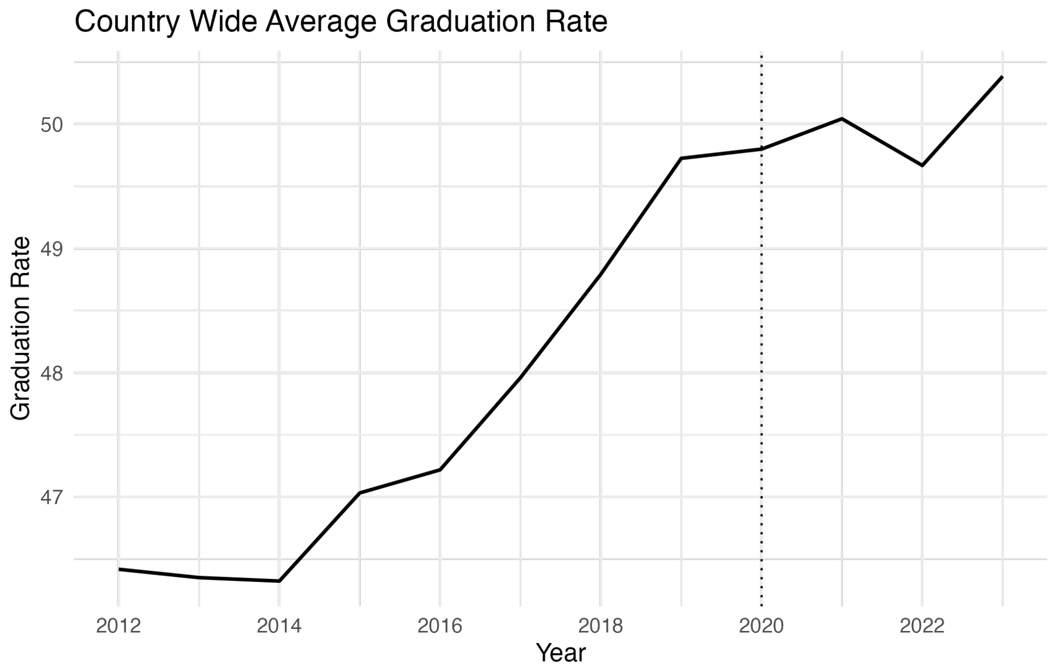


Figure 2: Average Graduation Rate, Oregon vs Other States

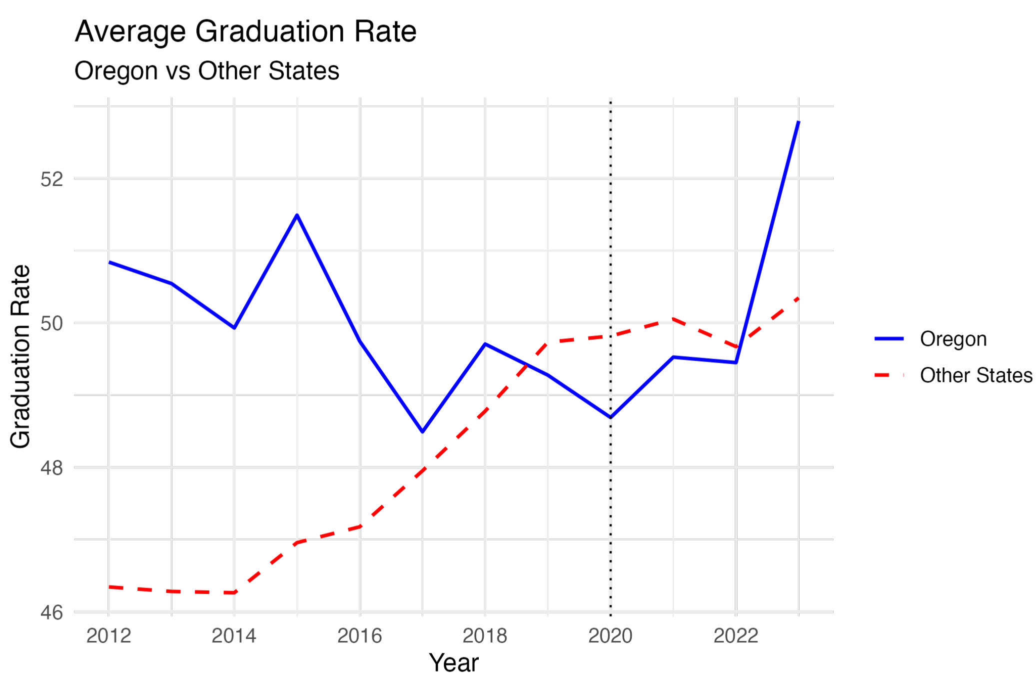


Figure 3: Average Graduation Rate in Public Universities

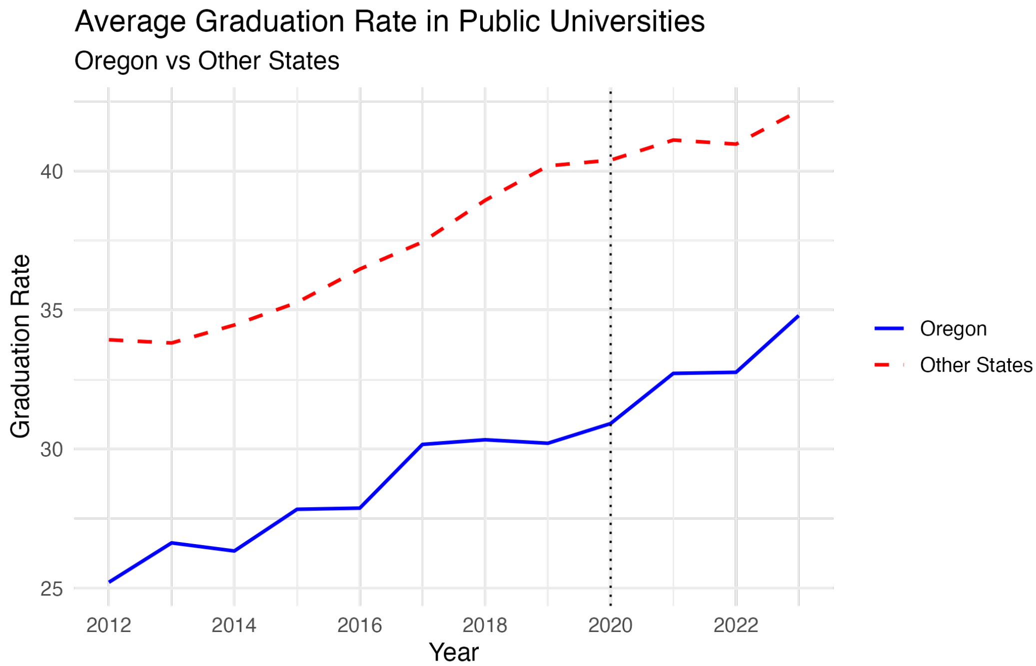
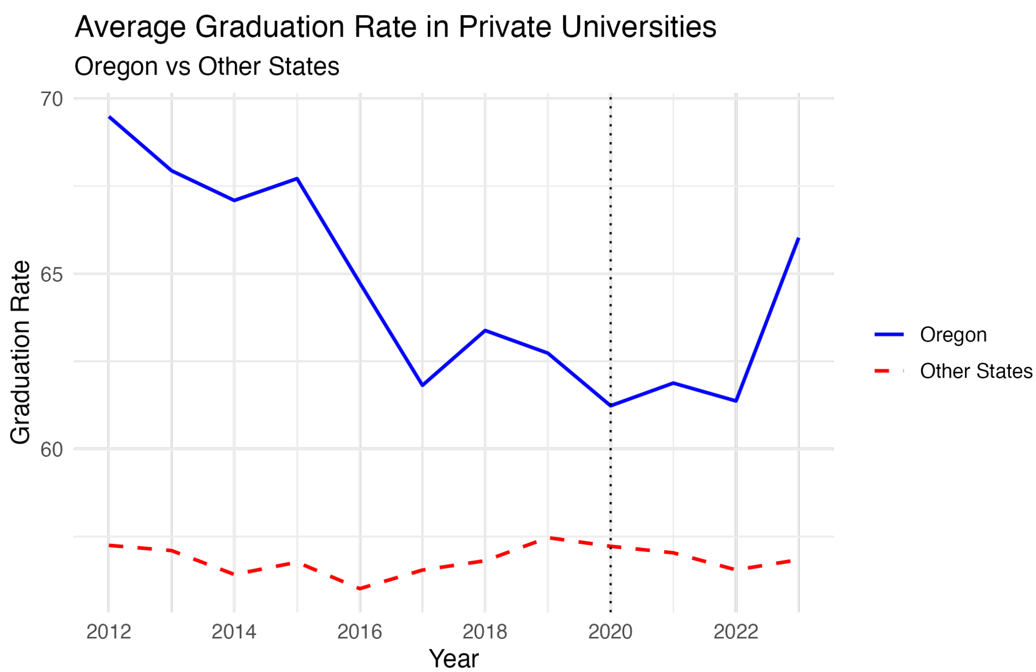


Figure 4: Average Graduation Rate in Private Universities



mizing the pre-treatment Mean Squared Prediction Error (MSPE) and ensuring an optimal counterfactual.

Let $Y_{\text{Oregon},t}$ be the observed outcome, the college graduation rate in Oregon at time t, and let $Y_{\text{synthetic},t}$ be the outcome for the "synthetic Oregon" at the same time t. The synthetic unit is a weighted average of the other states (the donor pool), with weights chosen to match Oregon's pre-treatment characteristics. Denoting the set of donor states as $j = 2, 3, \ldots, J$ and the corresponding optimal weights w_j^*, we can write:

$$Y_{\text{synthetic},t} = \sum_{j=2}^{J} w_j^* \, Y_{j,t} \tag{1}$$

Thus,

$$\text{ATT}_t = Y_{\text{Oregon},t} - Y_{\text{synthetic},t}. \tag{2}$$

Despite the advantages of employing a synthetic control approach, several challenges may arise in this setting. The effects of decriminalization on educational outcomes may take several years to manifest; a short, in this case three-year, post-treatment window might not fully capture Measure 110's long-term impact. Concurrent policy changes—such as alterations in financial aid, tuition costs, or other statewide reforms—can further complicate the isolation of Measure 110's effect, though I did not find any evidence of this in my research. Finally, constructing a valid synthetic counterfactual requires identifying a donor pool of states that credibly resembles Oregon's pre-treatment trends, which can be difficult if Oregon is an outlier or if no suitable combination of donor states exists—a violation of the convex hull assumption.

One of the most significant threats to identification is the effect of COVID-19 on graduation rates, which coincided with the implementation of Measure 110. Although the pandemic affected every state, local responses, severity, and the resulting impact on educational outcomes likely varied. Ideally, controlling for a time fixed effect and including an interaction term between COVID-19 severity and state would help mitigate these confounding effects.

To address this directly, the robustness section reports a Synthetic Difference-in-Differences estimator (Arkhangelsky, Athey, Hirshberg, Imbens, & Wager, 2021), which admits a common time shock through time weights and so differences out a nationwide pandemic effect rather than attributing it to the treatment, together with in-time placebo (backdating) tests that check whether the pandemic alone generates a spurious effect.

To credibly estimate a causal impact, several assumptions must hold. Most critically, the synthetic control method relies on the assumption that, in the absence of treatment, Oregon would have followed the same path as the composite of donor states. This requirement necessitates strong pre-treatment fit between Oregon and the synthetic counterpart, which is explored in my preliminary results. Another key assumption is that Measure 110 does not significantly alter outcomes in other states, since potential cross-border effects would compromise the donor states' validity. Nevada, a state bordering Oregon, in fact receives the largest weight in the main synthetic control, so I address this concern directly in the robustness section by re-estimating with all four bordering states (California, Idaho, Nevada, and Washington) removed from the donor pool. I am not worried about student migrations due to IPEDS' calculation of graduation rates being via cohort and controlling for transfers in and out. Though even if you did believe graduation rates were being affected by student migration because of Measure 110, this is still a valid effect within the ATT framework.

Additionally, since I have data at the institution level and we see that graduation rates differ across institution sectors, some heterogeneity analysis is warranted. To do this, I subsample across sector and level, meaning public vs private and urban vs non-urban institutions. Splitting the sample allows the treatment effects to be different across college types.

A final important piece of context in this analysis is that Measure 110 may have influenced graduation rates primarily through its impact on drug use, making any observed effect effectively a secondary or indirect policy outcome. Ideally, I would have directly measured changes in drug use following Measure 110 and then employed mediation analysis to determine the extent to which drug use explained shifts in graduation rates. Unfortunately, time constraints prevented me from conducting that additional layer of analysis.

6 Preliminary Results

In this section, I present my preliminary results and discuss their implications. I begin by reporting the main findings without subsampling the data and then explore potential heterogeneous treatment effects by splitting the dataset based on various institutional characteristics.

Figure 5 shows the synthetic control graph, comparing Oregon's graduation rates to those of its synthetic counterpart over time. A visual inspection reveals a strong pre-treatment fit, consistent across all subsequent subsamples and specifications. While a naïve interpretation might suggest that Measure 110 increased Oregon's graduation rate by 4 percentage points by 2023, in-space placebo tests do not support this conclusion. As shown in Table 2, the p-values for all estimates are insignificant, and Figure 6 further demonstrates that this outcome is not unique.

Table 2: ATTs and Associated P-Values, All Institutions

	Estimate	P-value	Standardized P-value
2020	-1.024	0.58	0.420
2021	1.492	0.44	0.320
2022	0.840	0.86	0.520
2023	4.053	0.16	0.160

All institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

For the rest of this section I show just the tables, like Table 2, for the remaining subsamples, none of which result in any significant effects.

Figure 5: Oregon vs Synthetic Oregon

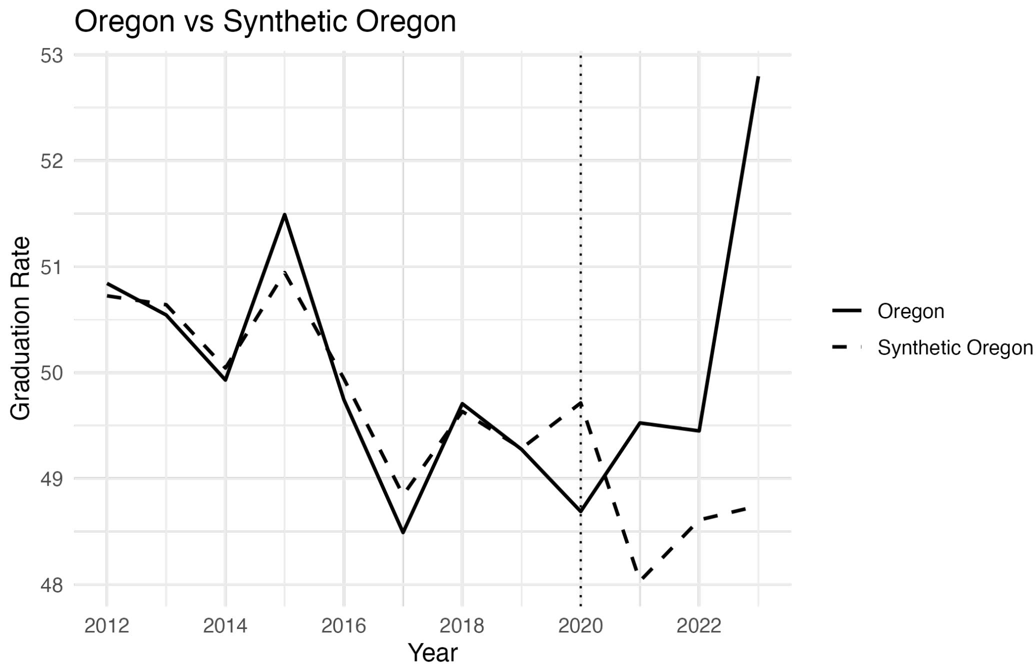


Figure 6: Placebo Tests of Oregon vs Other States

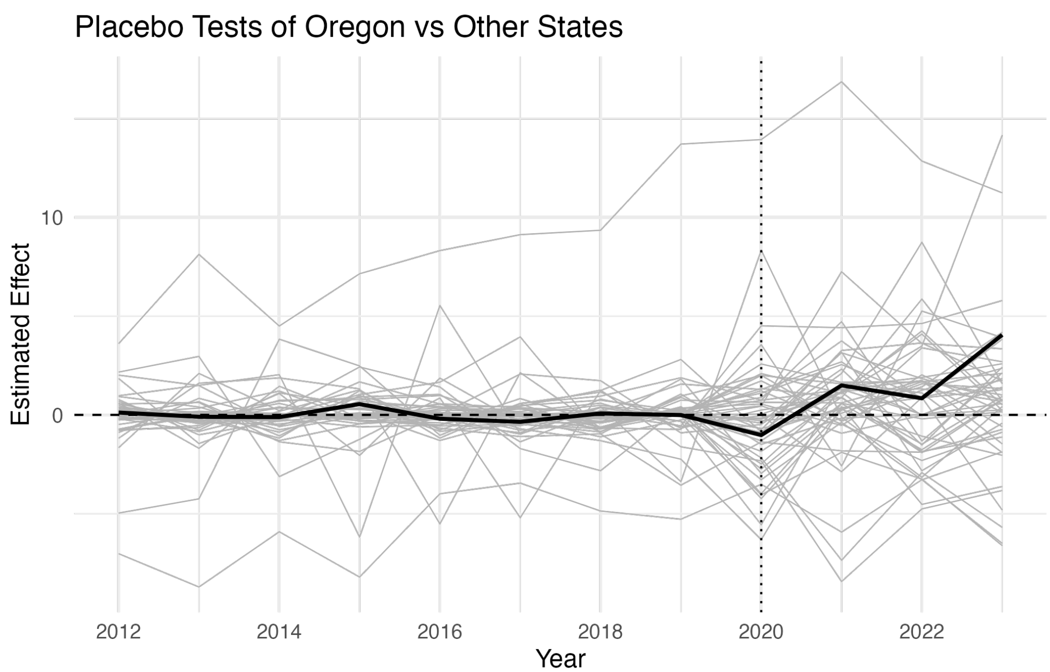


Table 3: ATTs and Associated P-Values, Public Institutions

	Estimate	P-value	Standardized P-value
2020	-0.149	0.94	0.940
2021	1.136	0.60	0.460
2022	0.083	0.92	0.880
2023	3.001	0.26	0.320

Public institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

Table 4: ATTs and Associated P-Values, Private Institutions

	Estimate	P-value	Standardized P-value
2020	-1.449	0.72	0.620
2021	1.741	0.64	0.620
2022	1.595	0.60	0.560
2023	7.709	0.10	0.140

Private institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

Table 5: ATTs and Associated P-Values, Urban Institutions

	Estimate	P-value	Standardized P-value
2020	-0.987	0.84	0.800
2021	-1.742	0.58	0.620
2022	-1.398	0.68	0.640
2023	3.870	0.44	0.520

Urban institutions. ATTs and p-values constructed via in-space placebo tests (R implementation of `synth_runner`).

7 Robustness

Because the headline result is a null, its credibility rests on showing that it survives reasonable perturbations of the design and that the two most serious confounds—the COVID-19 pandemic and contamination of the donor pool—do not overturn it. This section addresses each in turn and closes with specification and inference checks.

7.1 COVID-19 confounding

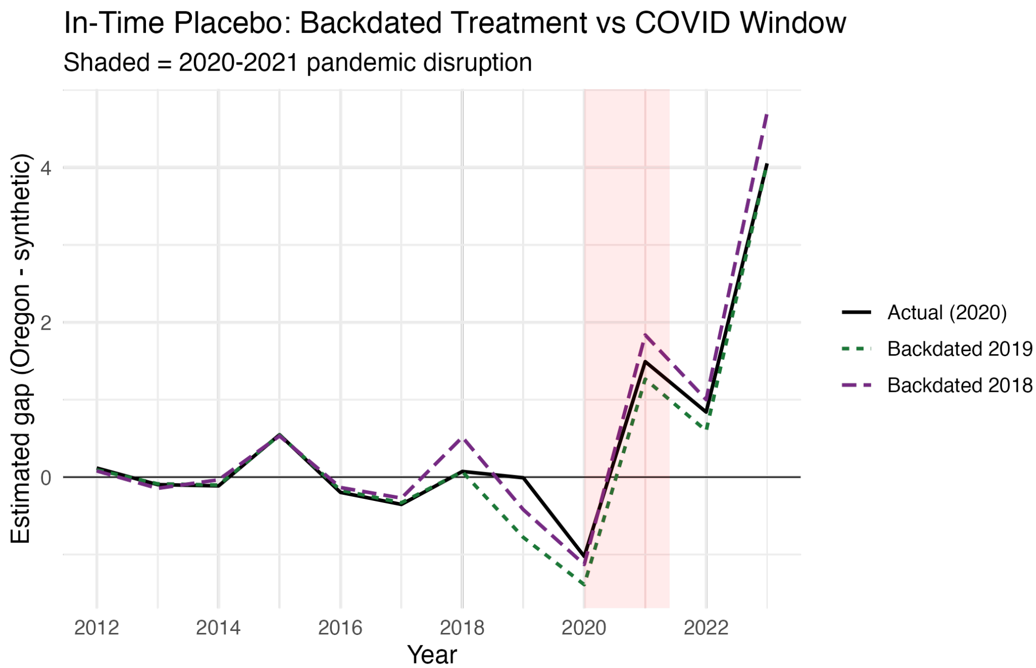
Measure 110 took effect in February 2021, inside the pandemic's disruption of higher education, so a natural worry is that any post-2020 gap between Oregon and its synthetic counterpart reflects state-specific pandemic effects rather than the policy. I address this in two ways. First, I re-estimate the effect using Synthetic Difference-in-Differences (Arkhangelsky et al., 2021), which augments the synthetic control with time weights and a difference-in- differences structure so that a common nationwide shock is differenced out rather than attributed to the treatment. Second, I run in-time placebo (backdating) tests: I assign a *fake* treatment year that precedes both the policy and the pandemic (2018 and 2019) and check whether a spurious "effect" appears once COVID-19 hits. As Figure 7 and Table 6 show, backdated treatments produce 2020–2021 gaps comparable in magnitude to Oregon's true post-2020 gap, indicating that the observed movement is largely pandemic-driven common variation rather than a Measure 110 effect. The SDID estimates in Table 8 tell the same story: the estimated ATT is small and its confidence interval includes zero in every specification.

Table 6: In-Time Placebo: Pandemic-Window Gaps by (Fake) Treatment Year

Treatment year	2020 gap	2021 gap
Actual (2020)	-1.02	1.49
Backdated 2019	-1.39	1.27
Backdated 2018	-1.13	1.84

Estimated Oregon-minus-synthetic gap in the 2020 and 2021 pandemic-window years under the actual 2020 treatment date and two pre-policy, pre-COVID fake treatment dates. Comparable gaps under the fake dates indicate the post-2020 movement reflects the pandemic rather than Measure 110.

Figure 7: In-Time Placebo Tests (Backdated Treatment)



7.2 Donor-pool contamination

Two threats concern the donor pool. First, cross-border spillovers: if Measure 110 changed behavior in neighboring states, those states are invalid controls. Because Nevada—an Oregon border state—receives the largest weight in the main fit, I re-estimate all four specifications with California, Idaho, Nevada, and Washington removed from the donor pool (Table 7). Doing so cuts the pooled 2023 estimate roughly in half, from 4.05 to 2.48 percentage points, and it remains statistically insignificant ($p = 0.24$); the other three subsamples are essentially unchanged. Second, a broader "harm-reduction wave": several donor states liberalized their own drug or recreational-marijuana laws during 2020–2023, which could bias the counterfactual. Re-estimating the pooled specification with those ten states removed leaves the 2023 estimate near its main value (4.24); its raw placebo p-value falls to 0.05, but the RMSPE-standardized p-value—less sensitive to the smaller placebo pool—remains insignificant at 0.13. Finally, a leave-one-out exercise (Figure 8) drops each positive-weight donor in turn: removing Nevada lowers the 2023 gap to 2.28 while every other donor leaves it between 3.9

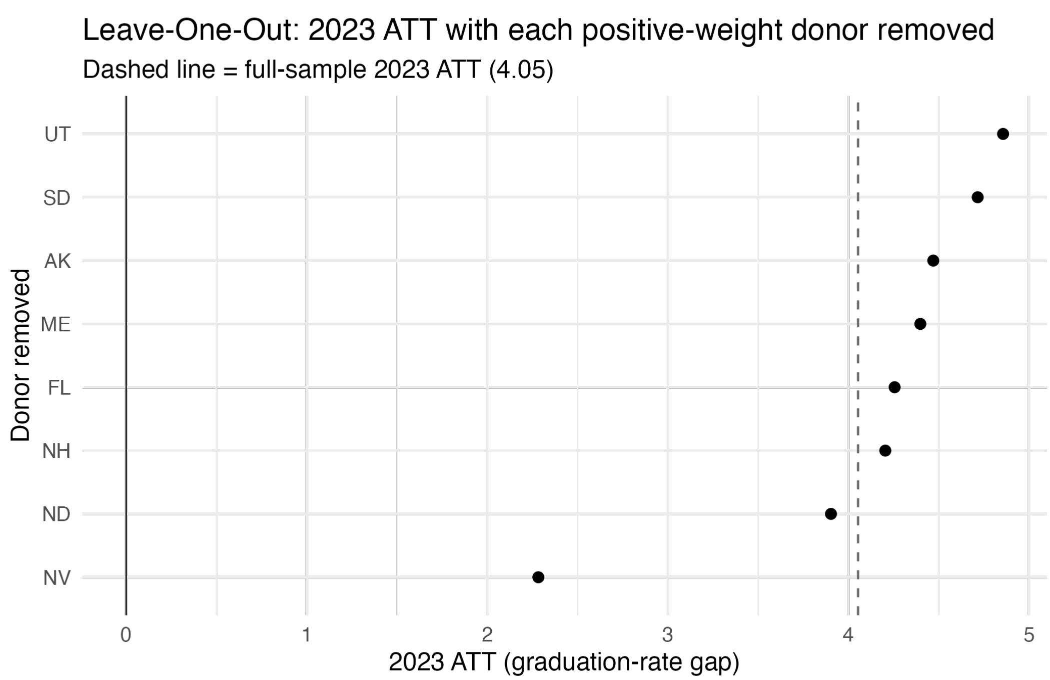
and 4.9, and no single exclusion makes the estimate significant (p between 0.14 and 0.33). The magnitude of the (insignificant) post-2020 gap is thus largely a Nevada artifact, and no donor-pool restriction overturns the null.

Table 7: ATTs with Border States Removed from the Donor Pool

	Estimate	P-value	Standardized P-value
2020	-0.849	0.65	0.652
2021	-0.600	0.63	0.630
2022	-0.497	0.89	0.848
2023	2.481	0.24	0.261

Pooled specification with California, Idaho, Nevada, and Washington excluded from the donor pool. Per-period ATTs and in-space placebo p-values.

Figure 8: Leave-One-Out: 2023 ATT with Each Positive-Weight Donor Removed



7.3 Alternative estimators, specification, and inference

Table 8 reports the Synthetic DiD and Augmented SCM (Ben-Michael, Feller, & Rothstein, 2021) estimates side by side. Augmented SCM ridge-corrects any residual pre-treatment imbalance and returns a confidence interval; that interval is informative about *power*, as it bounds the effect sizes the design could plausibly detect. The pooled average post-treatment ATT is 1.04 points under Synthetic DiD (95% CI $[-3.29, 5.38]$) and 1.45 points under Augmented SCM (95% CI $[-1.38, 3.54]$); every interval, in every subsample, includes zero. The width of these intervals is itself informative: with only three post-treatment years and a single treated state, the design can rule out large effects but not moderate ones, so the null should be read as "no evidence of a sizable effect" rather than "evidence of no effect."

Table 9 adds the Mean Squared Prediction Error ratio test (Abadie et al., 2010), an overall permutation p-value that ranks Oregon's post-to-pre MSPE ratio against the placebo distribution; Oregon is not an outlier in any subsample. Table 10 varies the specification: using three outcome lags plus covariates, covariates only, and an enrollment-weighted state collapse, as well as pinning the predictor-weight matrix V to Stata's regression-based heuristic rather than the nested optimum. The point estimate is itself fragile—the 2023 pooled gap ranges from slightly negative under the enrollment-weighted collapse to roughly five percentage points under a three-lag specification—but it is statistically insignificant throughout and is not even robust in sign. If anything, this fragility reinforces the central finding: there is no stable, detectable Measure 110 effect, and the headline magnitude is an artifact of the specific predictor set and V-selection method rather than a real treatment response. The V-selection comparison also documents the one material discrepancy between this analysis and the original Stata results: the nested optimization used by R's `Synth` and Stata's regression-based default produce slightly different weights and hence estimates (for example, a 2023 pooled ATT of 4.05 versus 4.22), while leaving every inferential conclusion unchanged.

Table 8: Alternative Estimators: Synthetic DiD and Augmented SCM

	Synthetic DiD ATT [95% CI]	Augmented SCM ATT [95% CI]
All institutions	1.04 [-3.29, 5.38]	1.45 [-1.38, 3.54]
Public institutions	0.51 [-4.80, 5.82]	1.02 [-6.05, 3.30]
Private institutions	0.73 [-5.39, 6.85]	2.31 [-1.85, 6.34]
Urban institutions	-1.07 [-6.97, 4.84]	-0.05 [-3.36, 4.44]

Average post-period ATT with 95% confidence intervals. Synthetic DiD standard errors are from the placebo method; Augmented SCM (ridge) intervals are jackknife. Every interval includes zero.

Table 9: MSPE-Ratio Permutation Test

Sample	MSPE ratio	Rank	P-value
All institutions	80.90	14/51	0.275
Public institutions	25.27	22/51	0.431
Private institutions	35.85	17/51	0.333
Urban institutions	10.38	35/51	0.686

Ratio of post-treatment to pre-treatment MSPE for Oregon, its rank among all units (treated plus placebos), and the rank-based p-value.

Table 10: Specification and V-Selection Sensitivity (Pooled ATT)

Specification	2021	2022	2023
All lags (main)	1.49	0.84	4.05
3 lags + covars	2.84	1.74	5.58
Covariates only	0.19	-1.48	3.16
Enrollment-weighted	-0.00	-0.11	-0.72
Stata heuristic V	1.58	0.90	4.22

Pooled per-year ATTs under alternative predictor sets, an enrollment-weighted state collapse, and Stata's regression-based V (rather than the nested optimum used in the main analysis).

8 Conclusion

I did not find any evidence that Measure 110 influenced college graduation rates under any of the specifications or subsamples tested. Future research may be warranted to first determine whether Measure 110 affects college students' drug use, as that could provide a clearer link between the policy and educational outcomes. However, based solely on the results here, there is no indication that Measure 110 has a significant impact on graduation rates. Exploring further subsampling and heterogeneity analyses across different demographic groups and institutional characteristics could be a promising direction for additional study.

References

Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of california's tobacco control program.

Journal of the American Statistical Association, *105*(490), 493–505.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., & Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, *111*(12), 4088–4118.

Arria, A. M., Caldeira, K. M., Bugbee, B. A., Vincent, K. B., & O'Grady, K. E. (2015). The academic consequences of marijuana use during college. *Psychology of Addictive Behaviors*, *29*(3), 564–575.

Ben-Michael, E., Feller, A., & Rothstein, J. (2021). The augmented synthetic control method. *Journal of the American Statistical Association*, *116*(536), 1789–1803.

Caldeira, K. M., Kasperski, S. J., Sharma, E., Vincent, K. B., O'Grady, K. E., Wish, E. D., & Arria, A. M. (2009). College students rarely seek help despite serious substance use problems. *Journal of Substance Abuse Treatment*, *37*(4), 368–378.

Chatterji, P. (2003). Illicit drug use and educational attainment. *Health Economics*, *12*(5), 377–391.

Kerr, D. C. R., Bae, H., & Koval, A. L. (2018). Oregon recreational marijuana legalization: Changes in undergraduates' marijuana use rates from 2008 to 2016. *Psychology of Addictive Behaviors*, *32*(6), 670–678.

Maghsoudi, N., McDonald, K., & Thompson, H. (2021). Drug checking services for people who use drugs: A systematic review. *Addiction*, *116*(2), 297–310.

Mezza, A., & Buchinsky, M. (2021). The effect of drug use on education and labor market outcomes: A dynamic structural approach. *Journal of Econometrics*, *220*(2), 474–493.

Spencer, N. (2023). Does drug decriminalization increase unintentional drug overdose deaths? Early evidence from Oregon Measure 110. *Journal of Health Economics*, *91*, 102798.

Welsh, J. W., Shentu, Y., & Sarvey, D. B. (2019). Substance use among college students. *Focus*, *17*(2), 117–127.

Wen, H., Hockenberry, J. M., & Cummings, J. R. (2015). The effect of medical marijuana laws on adolescent and adult use of marijuana, alcohol, and other substances. *Journal of Health Economics*, *42*, 64–80.

Zoorob, M. J., Park, J. N., Kral, A. H., Lambdin, B. H., & Del Pozo, B. (2024). Drug decriminalization, fentanyl, and fatal overdoses in Oregon. *JAMA Network Open*, *7*(9), e2431612.